Statistical methods

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May 2026

1 Project Links

GitHub Repository:
https://github.com/Valementajat/Statistical_methdos

Supervised data set:
https://archive-beta.ics.uci.edu/dataset/915/differentiated+thyroid+cancer+recurrence

Unsupervised data set:
https://archive-beta.ics.uci.edu/dataset/963/ur3+cobotops

2 Introduction

In this report for the course: Machine Learning e Statistical Learning - Module Statistical Learning. I will present two different projects in the R language: a supervised and an unsupervised one. I will go through what I have achieved and what steps I have taken to achieve the results.

The goal of the projects was to carry out a complete statistical analysis in R with the following criteria.

- Use a suitable cross-sectional dataset or process a dataset into a suitable format.
- Apply supervised and unsupervised methods can be to different data sets.
- Evaluate and interpret the results in relation to the research questions.
- Present the work clearly in a written report.

Both datasets used were retrieved from the UCI Machine Learning Repository. The datasets were last accessed on 3 May 2026.

In general, a complete analysis was achieved from these data sets.

3 Declaration

I/We declare that this material, which I/We now submit for assessment, is entirely my/our own work and has not been taken from the work of others, save and to the extent that such work has been cited and acknowledged within the text of my/our work. I/We understand that plagiarism, collusion, and copying are grave and serious offences in the university and accept the penalties that would be imposed should I engage in plagiarism, collusion, or copying. This assignment, or any part of it, has not been previously submitted by me/us or any other person for assessment on this or any other course of study.

4 Research Questions and Hypotheses

The project is divided into two parts: a supervised classification analysis and an unsupervised clustering analysis.

For the supervised analysis, the main research question is:

- **RQ1:** Which clinical variables appear to be associated with thyroid cancer recurrence?

The related hypothesis is:

- **H1:** Thyroid cancer recurrence is associated with clinical response and other clinical characteristics.

For the unsupervised analysis, the main research question is:

- **RQ2:** Can observations in the UR3 CobotOps dataset be grouped into meaningful clusters based on the available sensor and operational variables?

The related hypothesis is:

- **H2:** The UR3 CobotOps dataset contains underlying patterns that can be identified using clustering methods.

5 Datasets and Preprocessing

Both datasets used in the projects:

- Differentiated Thyroid Cancer Recurrence
- UR3 CobotOps

were retrieved from the UCI Machine Learning Repository at 3 May 2026.

5.1 Supervised

For the supervised classification task, The **Differentiated Thyroid Cancer Recurrence** dataset was used. The target variable was the **recurred** column, which indicates whether recurrence occurred, and the data was divided into training and testing sets. No extensive pre-processing required.

5.2 Unsupervised

For the unsupervised clustering task, the **UR3 CobotOps** dataset was used. Since the data were time-series, it had to be transformed into a cross-sectional format to comply with the project requirements. This was done by aggregating the time-series measurements so that each row represented one independent observation suitable for clustering. After this transformation, the selected numerical variables were scaled before applying clustering methods, because variables measured on different scales can otherwise dominate the clustering results.

6 Supervised analysis

For the supervised analysis, the process was quite straightforward because the dataset already had a clear target variable: `recurred`, which tells whether thyroid cancer recurrence happened or not.

For the research question 1, I wanted to find out which clinical variables appear to be associated with thyroid cancer recurrence. To do this, I used variables such as age, gender, risk, stage, response, and adenopathy as predictors.

The first step I took was to explore the dataset with summary statistics and plots. Then I used tests to check whether some variables seemed to be related to recurrence. After that, I trained two classification models: logistic regression and a decision tree. Finally, I compared their performance using accuracy, sensitivity, specificity, and balanced accuracy.

6.1 Exploratory analysis

Overall notices from the data analysis:

- There are no missing values in the dataset
- The dataset contained 383 patients, and the majority of them were female
- The target variable was also somewhat imbalanced, as 275 patients from 383 had no recurrence
- Most patients had papillary thyroid cancer
- The majority of patients were classified as being low risk

During the exploratory analysis, some simple plots were created. The goal of these plots were to get a deeper understanding of the target variables' association to some clinical variables.

First, I plotted the distribution of the target `recurred` variable as it's useful for checking the class balance between patients who had recurrence and patients who did not.

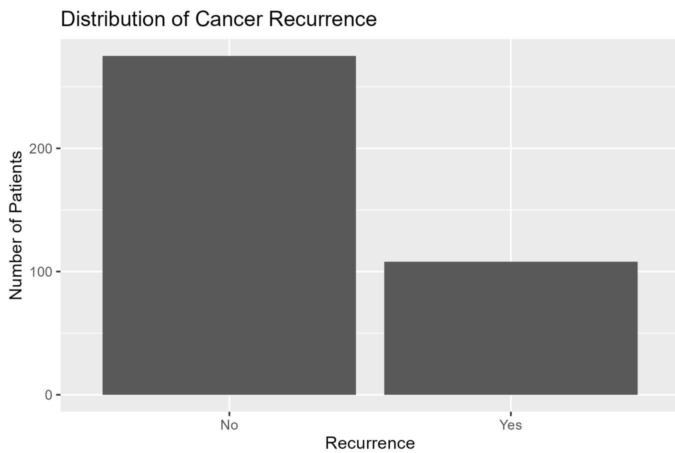

Figure 1: Distribution of the recurrence target variable

Next, I plotted some of the categorical values which in my mind made sense with the target variable, like the recurrence proportion against the patients response to treatment.

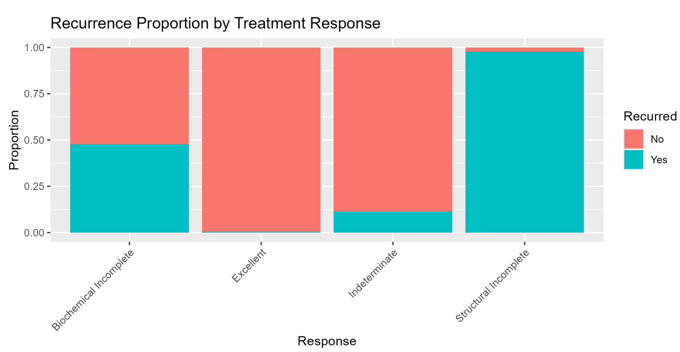

Figure 2: Distribution of treatment response against recurrence

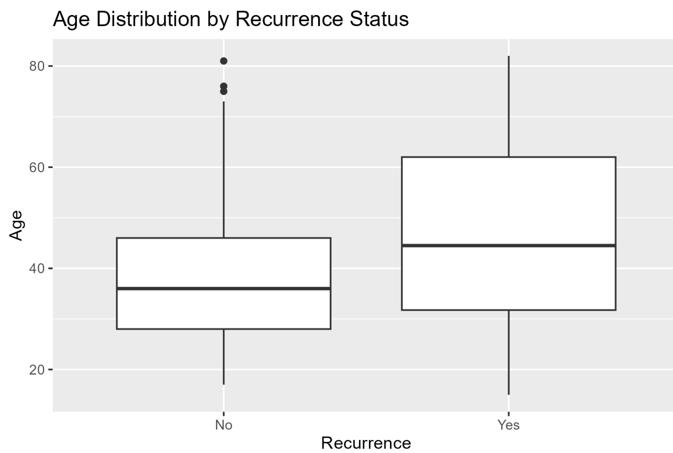

Figure 3: Distribution of age against recurrence

Although these plots do not serve as the sole truth to prove causality, they serve as an indication that, in the case of these plots, age and treatment response might contain useful information for predicting the recurrence of cancer.

Only a selection of the exploratory plots is shown here, focusing on variables that appeared most relevant to the task of supervised classification.

6.2 Statistical association tests

To accompany the exploratory analysis, association tests were run to see whether recurrence was related to the other variables. Due to the dataset being mainly categorical, a Fisher's exact test was implemented to find associations between categorical variables. Age being the only numerical variable, for that test, the recurrence was mutated into 1 or 0 numbers. Because some variables had several categories, simulated p-values were used to avoid computational limitations of the exact calculation.

In Table 1, the *Statistic* column reports the test statistic, which is only available for age. For the Spearman correlation test, this is the correlation coefficient ρ, which is the direction and strength of the relationship between age and recurrence. A positive value means that recurrence tends to be more common as age increases. For Fisher's exact test, no single test statistic is reported in this table, as it only to see whether two categorical variables are associated, so the value is marked as NA.

The *p-value* column shows the probability of observing an association at least as strong as the one found in the data, assuming that there is no true association between the variable and recurrence. A small p-value, commonly below 0.05, suggests that the variable is statistically significantly associated with recurrence.

Table 1: Statistical association tests between variables and recurrence

Variable	Test used	Statistic	p-value
Age	Spearman correlation	0.212	< 0.001
Response	Fisher's exact test	NA	< 0.001
Risk	Fisher's exact test	NA	< 0.001
Stage	Fisher's exact test	NA	< 0.001
Adenopathy	Fisher's exact test	NA	< 0.001
Physical examination	Fisher's exact test	NA	0.007
Gender	Fisher's exact test	NA	< 0.001
Smoking	Fisher's exact test	NA	< 0.001
History of smoking	Fisher's exact test	NA	0.014
History of radiotherapy	Fisher's exact test	NA	0.002
Thyroid function	Fisher's exact test	NA	0.328
Pathology	Fisher's exact test	NA	< 0.001
Focality	Fisher's exact test	NA	< 0.001
T classification	Fisher's exact test	NA	< 0.001
N classification	Fisher's exact test	NA	< 0.001
M classification	Fisher's exact test	NA	< 0.001

From these tests, we can assume that every variable except **Thyroid function** is significant for predicting recurrence.

6.3 Modeling methods

Two types of models were tried for this dataset: a simple logistic regression and a decision tree model. Two models were chosen for comparison of the results.

6.3.1 Logistic Regression

At the start, a logistic regression model with all associated variables was considered, but after the first results showed abnormally out-of-place coefficients with huge standard errors, a culling of predictors was required.

Table 2: Selected logistic regression results for recurrence prediction

Variable	Estimate	Std. error	Statistic	p-value
Age	0.041	1613.367	0.000	1.000
Gender: Male	54.915	35303.677	0.002	0.999
Risk: Intermediate	79.114	118360.338	0.001	0.999
Risk: Low	42.481	122933.541	0.000	1.000
Stage II	144.388	47586.758	0.003	0.998
Response: Excellent	-275.325	57793.242	-0.005	0.996
Response: Indeterminate	-158.471	37968.714	-0.004	0.997
Response: Structural incomplete	125.294	69326.890	0.002	0.999
Adenopathy: Left	106.611	187794.593	0.001	1.000
Adenopathy: Right	94.223	100326.654	0.001	0.999
Smoking: Yes	67.925	87958.278	0.001	0.999
History of smoking: Yes	-8.178	150944.004	0.000	1.000

In the reduced model, due to almost all variables having a strong association to recurrence and the initial prediction having extreme standard errors, the predicting variables were chosen for their following qualities.

- Age: Baseline patient-level information
- Gender: Baseline patient-level information
- Response : Summarizes clinically important information about disease status and projection
- risk : Summarizes clinically important information about disease status and projection
- Adenopathy : Summarizes how the body's immune system responds to the diseases

Table 3: Reduced Logistic regression results for recurrence prediction

Term	Estimate	Std. error	p-value
(Intercept)	14.399	4898.582	0.998
Age	0.054	0.037	0.140
Gender: M	1.044	0.869	0.230
Risk: Intermediate	-17.458	4898.582	0.997
Risk: Low	-21.045	4898.582	0.997
Response: Excellent	-20.390	2016.374	0.992
Response: Indeterminate	-2.676	1.088	0.014
Response: Structural Incomplete	5.074	1.492	0.001
Adenopathy: Extensive	16.386	11191.689	0.999
Adenopathy: Left	0.827	3.279	0.801
Adenopathy: No	2.082	1.503	0.166
Adenopathy: Posterior	-0.358	21037.924	1.000
Adenopathy: Right	1.976	1.717	0.250

The reduced logistic regression model was easier to interpret than the full model. There were still some extremely huge standard errors. The next variable to be removed was risk due to its high p-value and std. error.

Table 4: Final Logistic regression results for recurrence prediction

Term	Estimate	Std. error	p-value
(Intercept)	-1.131	1.606	0.481
Age	0.037	0.021	0.086
Gender: M	0.416	0.766	0.587
Response: Excellent	-4.680	1.126	< 0.001
Response: Indeterminate	-2.525	0.745	< 0.001
Response: Structural Incomplete	2.947	0.927	0.001
Adenopathy: Extensive	14.079	1565.142	0.993
Adenopathy: Left	1.320	1.478	0.372
Adenopathy: No	-0.972	1.032	0.346
Adenopathy: Posterior	14.061	2779.686	0.996
Adenopathy: Right	0.515	1.208	0.670

6.3.2 Decision Tree

For the decision tree, I used the same reduced set of variables as in the logistic regression model. This was done to examine how the selected predictors split patients into recurrence and non-recurrence groups.

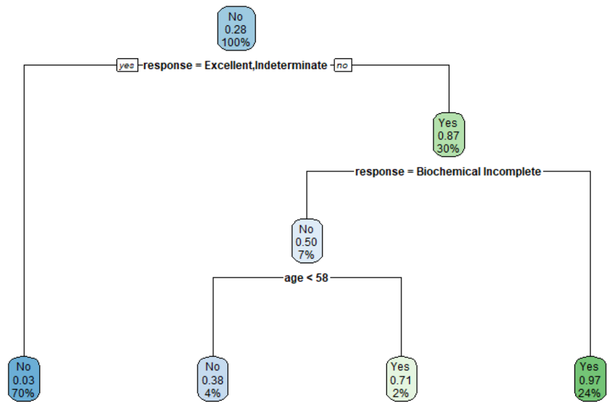

Figure 4: Decision Tree

The decision tree shows that treatment response was the most important splitting variable. The first split separates patients with an excellent or indeterminate response from the other response categories. Patients with excellent or indeterminate response were mostly classified as not having recurrence. On the other side of the tree, patients with incomplete structural responses were classified primarily as cases of recurrence.

6.4 Model comparison

The models were then compared using the same test set. The evaluation metrics were accuracy, sensitivity, specificity, and balanced accuracy. Accuracy shows the overall proportion of correct predictions. Sensitivity measures how well the model identifies recurrence cases, while specificity measures how well it identifies non-recurrence cases. Balanced accuracy was included because it gives a more balanced view when the classes are not perfectly equal.

Table 5: Model comparison on the test set

Model	Accuracy	Sensitivity	Specificity	Balanced accuracy
Logistic Regression	0.974	0.905	1.000	0.952
Decision Tree	0.961	0.857	1.000	0.929

Overall, both models performed well, but the reduced logistic regression model had the best balanced accuracy and is more suitable of the two.

6.5 Results and thoughts

Returning to the research question of which of these clinical variables are significant to predict recurrence and the hypothesis of it being associated with response, and other clinical characteristics. The results clearly indicate that from these clinical variabless, the most impactful variable is **response**, followed by risk, stage, and adenopathy. While these variables helped, the logistic regression model's coefficients showed them to be less stable or less clearly significant in the reduced model.

Overall, the supervised analysis supports the hypothesis that thyroid cancer recurrence is associated with response and other clinical characteristics. The strongest evidence was found for the treatment response, while the others had support behind them but should be interpreted with more caution.

7 Unsupervised Analysis

With unsupervised analysis, for originality's sake, I chose a time series dataset which required a lot of pre-processing to transform it to a suitable form. The goal of unsupervised analysis was to see if the grouped into meaningful clusters based on the available information.

The variables **robot_protective_stop** and **grip_lost** were not used for clustering. Instead, they were kept only for post-hoc interpretation, meaning that after the clusters were created, I checked whether the clusters were associated with these events.

7.1 Dataset processing

Now, as the selected dataset was a time series, it required hefty processing to be suitable for analysis. The original data contains timestamped sensor readings from robot operations, which means that there are multiple rows for the same robot cycle.

For the clustering and dimensionality reduction methods, the data were processed to represent one robot operation cycle. Because of this, the dataset was grouped by the variable **cycle**. After grouping, the sensor variables were aggregated using summary statistics. For each selected sensor variable, the mean, standard deviation, minimum, and maximum values were calculated. The idea of these is that the statistics act as the behavior of the sensor during one full cycle. Variables like **robot_protective_stop** and **grip_lost** were removed from the dataset and used as a post-hoc analysis to see if there are any meaninful clusters.

Before the data could be clustered, it had to be scaled using R's **scale()** function to avoid variables like temperature from being a dominant factor in clustering.

As the processed dataset still contained many sensor-derived variables, UMAP was applied to project the data into two dimensions.

7.2 Exploratory analysis

UMAP was used mainly as a visualization tool, allowing the high-dimensional sensor data to be inspected in a two-dimensional space.

Using the UMAP plots, we can explore whether the processed robot cycles showed any visible structure in the sensor data.

The first UMAP plot showed clear observations of clusters, and that the data it is not just randomly spread around.

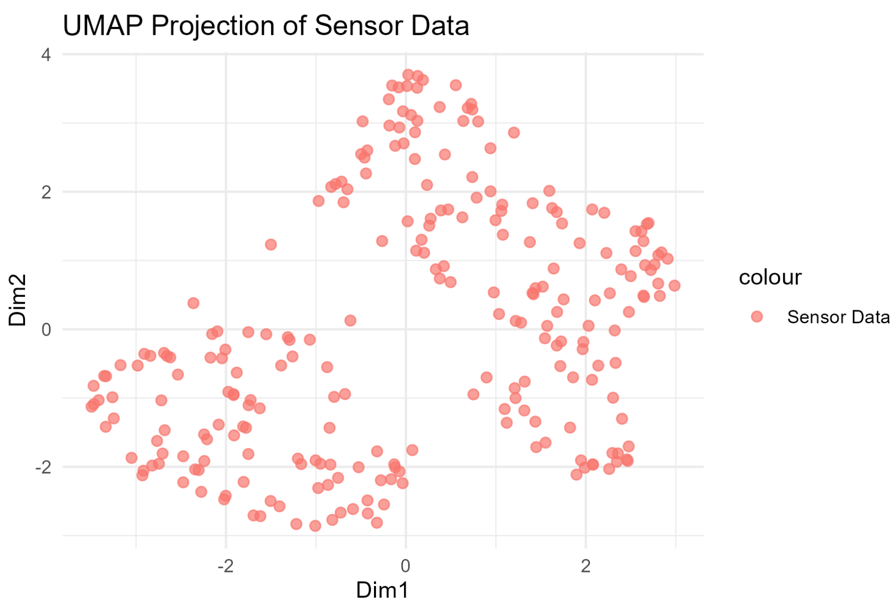

Figure 5: UMAP projection plot

The plot colored with a protective stop shows a clear grouping, especially in the right side, which suggests that the aggregated sensor variables contain patterns that separates different operation cycles.

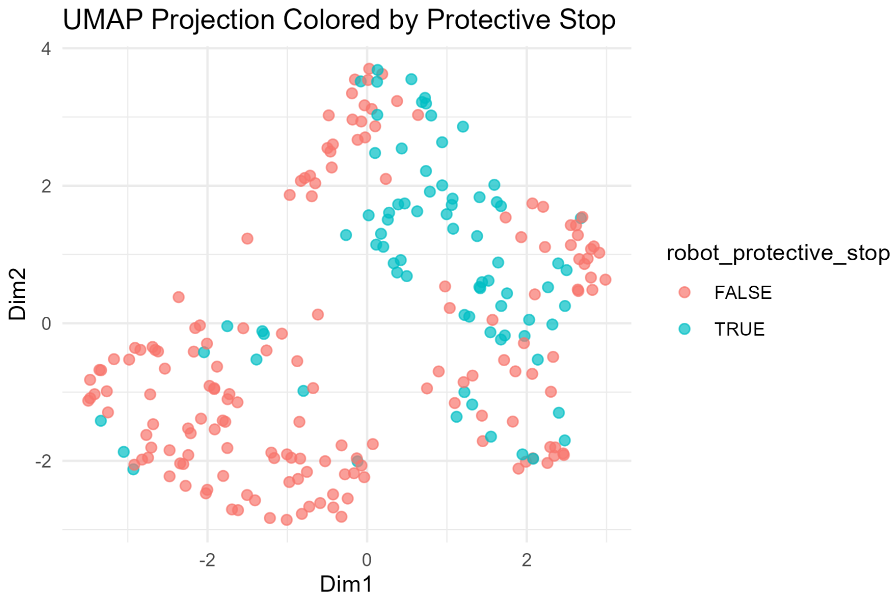

Figure 6: UMAP protective stop plot

Plot colored with grip loss showed weaker promise. Even if the plot shows some loose groups, these groups are spread around in several different regions. This means grip loss is harder to separate using only aggregated sensor data, or grip loss can appear under several conditions.

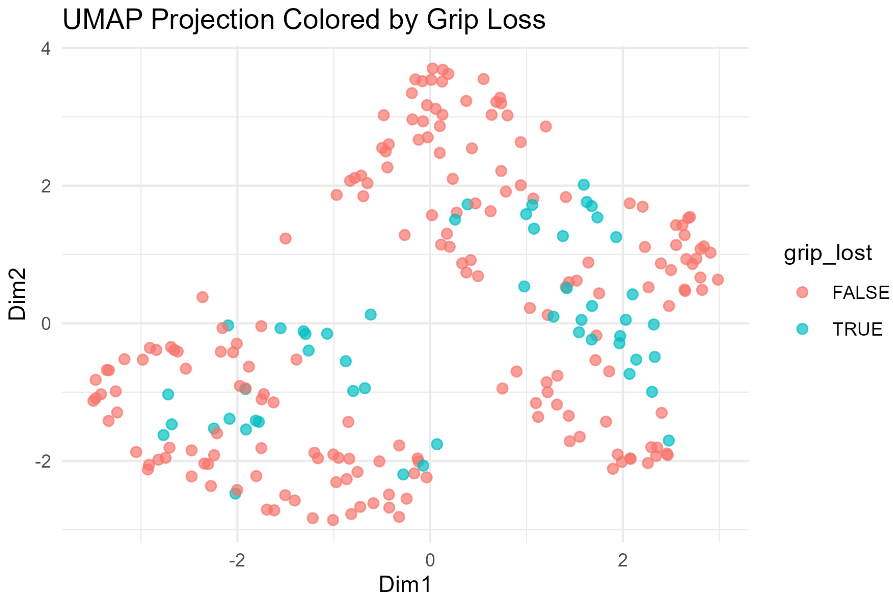

Figure 7: UMAP grip loss plot

7.3 Clustering with Gaussian Mixture Models

Gaussian mixture models were used from **mclust::Mclust** to identify similar groups with similar sensor behavior. The plotted model clearly shows that the model divided the UMAP projection into several clusters.

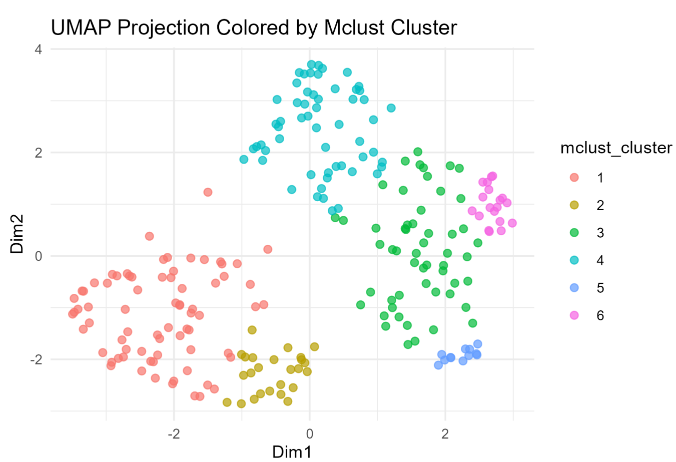

Figure 8: Gaussian Mixture Models Plot

After the Gaussian Mixture Model clusters were created, the clusters were compared with the two event labels: **robot_protective_stop** and **grip_lost**.

for both event labels, the number of robot cycles, event count, and rate were calculated for each cluster. This makes it possible to see whether some clusters represent more problematic robot behaviour than others.

Cluster	Cycles	Protective stops	Protective stop rate
3	54	31	57.4%
4	57	31	54.4%
5	12	3	25.0%
6	19	3	15.8%
1	73	9	12.3%
2	25	1	4.0%

Table 6: Protective stop rate by Mclust cluster.

Table 6 shows that protective stop events are not evenly distributed between clusters. Clusters 3 and 4 clearly have the highest protective stop rates, with approximately 57.4% and 54.4% of their cycles containing a protective stop.

Cluster	Cycles	Grip losses	Grip loss rate
3	54	23	42.6%
1	73	20	27.4%
2	25	3	12.0%
5	12	1	8.3%
4	57	4	7.0%
6	19	0	0.0%

Table 7: Grip loss rate by Mclust cluster.

Table 7 shows that grip loss also shows an uneven distribution, where the Cluster 3 has the highest loss rate with 42.6%, and it's being followed by the cluster 1 at 27.4%.

To compare these two event clusters, we can say that they are related to different sensor patterns. This can be seen where cluster 3 appears to be the most problematic cluster overall, because it has both a high protective stop rate and the highest grip loss rate. Cluster 4, on the other hand, seems more strongly associated with protective stops than with grip loss.

Overall, this post-hoc interpretation shows that the unsupervised clusters are not random. Although the event labels were not used for clustering, some clusters contain much higher proportions of protective stops or grip loss events than others.

7.4 Results and thoughts

Returning to the research question of can the UR3 CobotOps dataset be grouped into a meaningfull clusters and the hypothesis of it containing underlying patterns that can be used identified using clustering methods.

The Gaussian Mixture Model clustering does support this theory by dividing the UMAP projection into several clusters. This is further reinforced with the post-hoc analysis where event labels were used to interpret that some clusters were clearly more associated with problematic robot behavior than others.

8 Conclusion

Both supervised and unsupervised analyses were successful in relation to their individual research questions.

In the supervised analysis, the aim was to investigate which clinical variables were useful for predicting thyroid cancer recurrence. The results showed that variables such as treatment response, risk, stage, and adenopathy were strongly associated with recurrence.

In the unsupervised analysis, the aim was to investigate whether the UR3 CobotOps dataset contained meaningful groups of robot operation cycles. The results showed that the robot cycles did form visible groups, and post-hoc analysis showed some clusters had higher rates of protective stops or grip loss than others.

9 Use of AI

Due to English not being my first language, tools like Grammarly were used to check for grammar mistakes and reword some sentences to make them more coherent.

The use of AI tools for the project code itself was used to give examples and broaden the overall scope, an example of this is how the exploratory analysis was conducted.